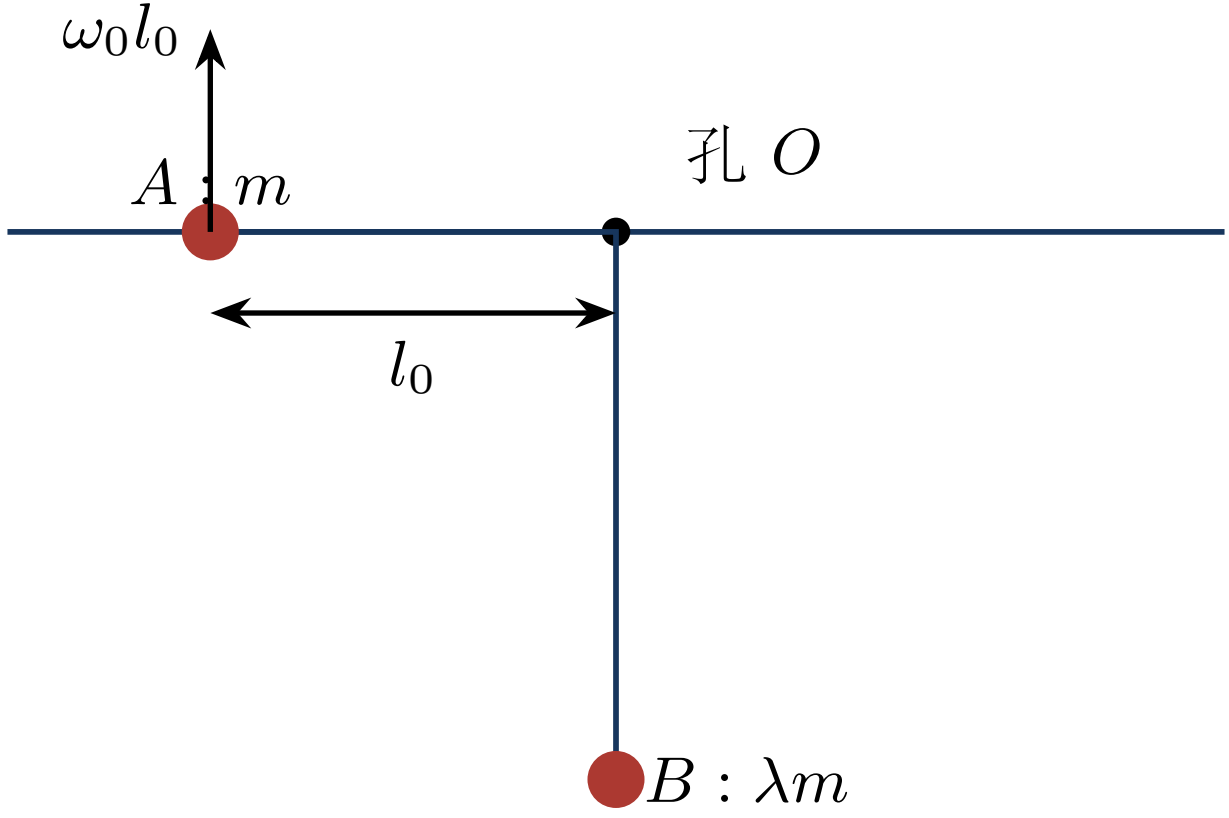
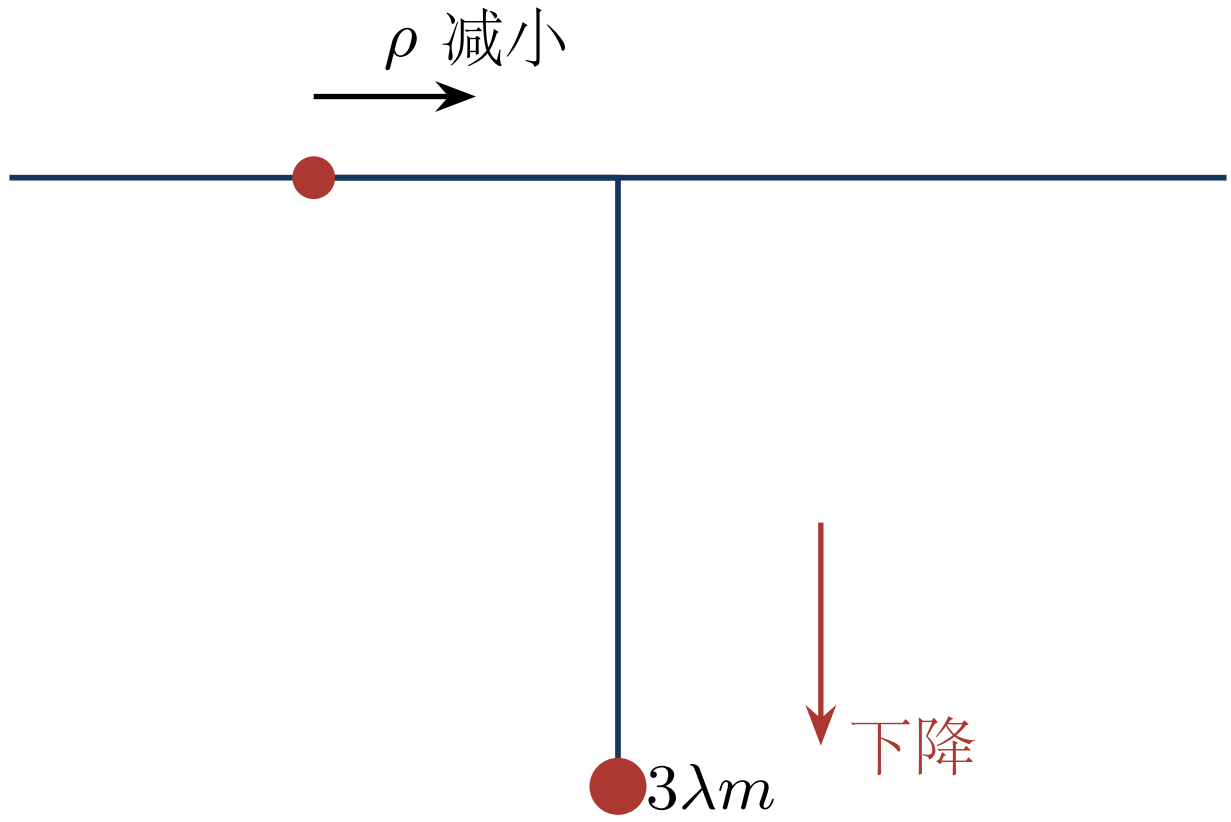
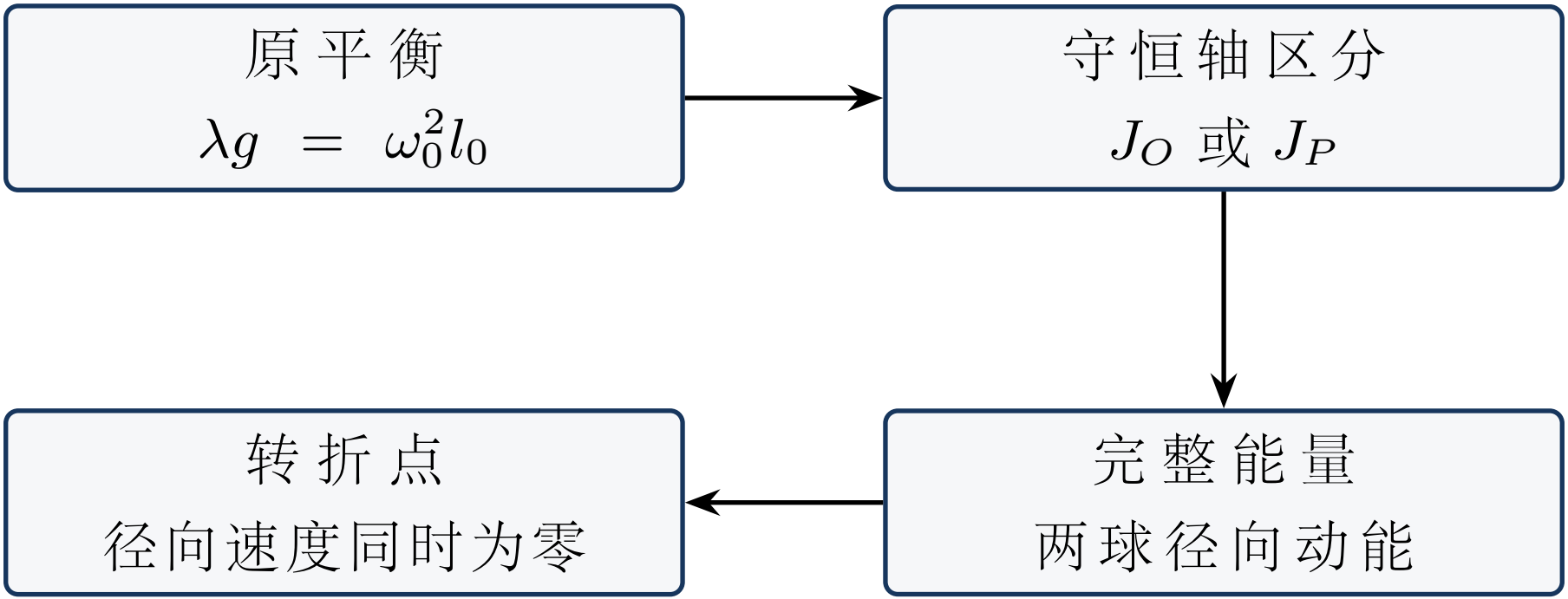


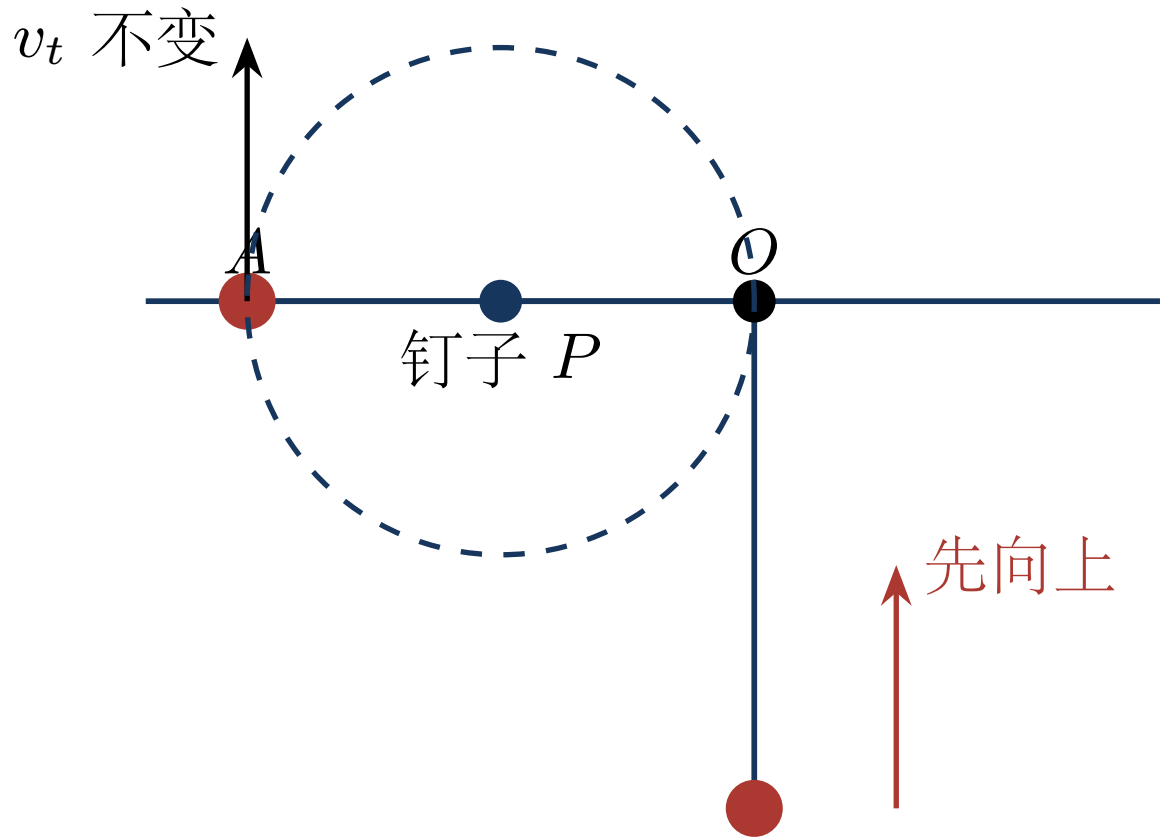
初态 $\lambda g \quad = \quad \omega_0^2 l_0$，下球静止



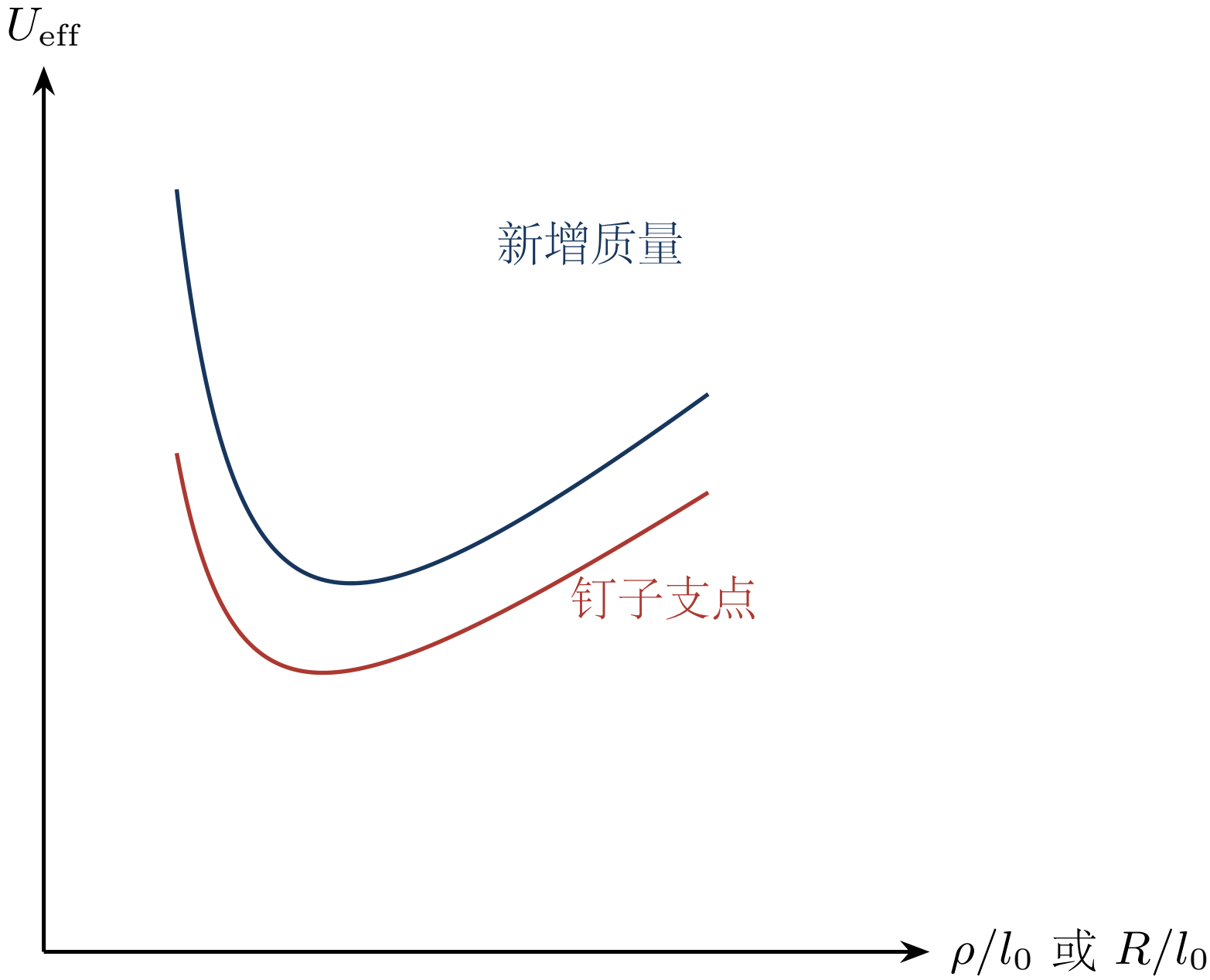
新增重物：旧转轴 O 不变，角动量守恒，切向速度增大



最大位移可精确求；大质量近似主要影响时间尺度



新半径 $R_0 = l_0/2$，新角动量 $J_P = mv_tR_0$



蓝：最近半径 $l_0/2$；红：最大上升 $(\sqrt{5}-1)l_0/4$

两曲线仅示各自有效势形状，归一化系数不同